Cersidy
My contribution to this lab was i was able to compare two models one for a singular decision tree and the other for the ensemble model I used the randomforestclassifier which used the wisdom of the crowd to make decisions while on this trial the difference wasn't much since i was using the iris dataset. My conclusion is that ensemble perform better because they give an average of many trees in this scenario to make a decision.

Samuel Camacho

Contribution

I did all the steps for the Module 09 lab on my own. I fixed the code errors, trained the Decision Tree and Random Forest models, created the feature importance chart, and answered the three questions at the end. All of my finished work is uploaded to my public GitHub (samcamacho-dotcom).

Ebuka

I worked independently to review the ensemble methods lab and the steps for classifying flowers using the Iris dataset. I examined the provided setup for splitting the data into training and testing sets and reviewed how to train and evaluate a Decision Tree and a Random Forest. I also reviewed the instructions for creating a feature importance plot and answering the knowledge check questions. This preparation helped me understand the code needed to complete the model comparison and interpret the results.

Sadia Ravate

For the Module 09 Ensemble Methods lab, I worked through the machine learning workflow using the Iris dataset. My contribution focused on completing the notebook, training and evaluating models, interpreting the results, and documenting my understanding of ensemble learning.

1. Notebook Setup and Data Preparation

I worked through the setup section by importing the required Python libraries and loading the Iris dataset. I used `train_test_split`to divide the data into training and testing sets. This helped me understand how data is prepared before model training.

2. Decision Tree Implementation

I created and trained a Decision Tree classifier using `DecisionTreeClassifier`. I used the training data to fit the model, generated predictions using the testing data, and evaluated the model with `accuracy_score`. The Decision Tree achieved 100% accuracy in my experiment.

3. Random Forest Implementation

I created a Random Forest classifier using 100 estimators. I trained the model, generated predictions, and calculated its accuracy. I also compared the Random Forest result with the Decision Tree to understand how ensemble methods perform relative to individual models.

4. Feature Importance Visualization

I extracted the feature importance values from the trained Random Forest and created a horizontal bar chart using Matplotlib. This visualization helped me understand how different flower measurements contributed to the model's classification decisions.

5. Knowledge Check and Documentation

I completed the knowledge check by explaining ensemble methods, comparing model performance, and interpreting the feature importance visualization. I also reviewed the lab concepts and prepared my reflective journal.

Through this lab, I practiced Python programming, data preparation, machine learning model training, model evaluation, and data visualization. I also strengthened my understanding of the difference between individual models and ensemble methods.

One challenge I encountered was a syntax error when the setup code was pasted without proper line breaks. I corrected the formatting and successfully executed the code. This reminded me that proper code formatting is important for Python execution.

Overall, my contribution involved completing the notebook workflow, evaluating both machine learning models, creating the feature importance visualization, and reflecting on the concepts learned. This lab improved my understanding of ensemble learning and gave me practical experience with machine learning in Google Colab